Velamala Pavan Krishna

+91 7569637875 | velamalapavankrishna@gmail.com | linkedin.com/in/pavan-krishna-velamala | github.com/W4RG0Dpk

Summary

Computer Science and AI undergraduate with a strong foundation in machine learning, deep learning, LLM-based systems, and data analysis. Hands-on experience in designing and implementing AI models for real-world applications. Highly motivated to learn, collaborate, and contribute to impactful AI-driven solutions.

Education

Amrita Vishwa Vidyapeetham — Kollam, Kerala
*B-Tech in Computer Science and Artificial Intelligence — **9.19 CGPA*** — *2023-2027*

Experience

Software Developer Intern — April 2025 - June 2025
Hunt Jobs Pvt Ltd Company — *Mumbai, Maharashtra*

- Built a modular, scalable resume parsing pipeline using Tesseract OCR for text extraction and LLM for context-aware entity recognition.
- Implemented a RAG based approach to enhance domain-specific NLP accuracy and speed in extracting skills, experience, and job titles while Embedding them in Faiss vector database
- Delivered an on-premise, high-performance system that significantly improved resume data processing accuracy, scalability, and efficiency.

HealthCare AI Club Member — April 2025 – Aug 2025
Amrita Vishwa Vidyapeetham — *Kollam , Kerala*

- This project is designed to detect Chagas disease from 12-lead ECG signals using a deep learning model (ResNet1D) combined with clinical signal-based features. It includes preprocessing, feature extraction, training, and inference modules.

Projects

Lumbar Spine Degenerative Classification Using Deep Learning| *Pytorch, Medical Imaging* — Sept-Dec 2025

- Designed a Multi Stage Deep Learning pipeline using a 2.5D MRI stacks input strategy for a UNet with SCSE attetion and EfficientNet-B0 encoder to localize the ROI from saggital T2 MRI and a slice-fusion EfficientNet-B0 classifier to predict the Disease.
- Achieved 91% accuracy and 73% F1 score when the outputs of localizer and classifier are given to a UNet Guided ROI Cropping Mechanism which is given as input to a EfficientNet-B0 classifier which grades the severity, Thus Achieving a Dedicated Severity Classification model.

Amazon Platform Chatbot | *Python, Fastapi, LLM ,RAG , Faiss* — Sept 2025

- Built a local RAG chatbot using MiniLM embeddings, FAISS vector search, and Mistral 7B LLM via Ollama for context-aware Amazon buyer and seller support.
- Developed FastAPI backend with semantic search, dynamic prompt generation, strict answer verification, and fallback handling to minimize hallucinations, ensuring trustworthy, context-grounded, precise responses with integrated user feedback and support mechanisms.

Disease Prediction based on symptoms | *Python, Machine Learning, Deep Learning* — April - July 2025

- Built a multi-class disease prediction pipeline using chi-square feature selection, MLP-based deep features, and synthetic data augmentation with VAEs, CVAEs, and SMOTE on a dataset of 773 diseases and 377 symptoms.
- Evaluated multiple classifiers including Logistic Regression, SVM, KNN, Random Forest, and XGBoost, achieving a peak accuracy of 89% with Logistic Regression on balanced data.

Centralized-DNS-with-Proxy-DNS | *Python, socket programming, Flask, SQLite, HTML* — Dec 2024

- Developed a centralized DNS system with proxy DNS server using Python socket programming, supporting load distribution, hostname and mail aliasing, and optimized caching to reduce DNS resolution latency.
- Designed a Flask web client for dynamic domain management, demonstrating strong networking fundamentals and scalable DNS architecture integration.

Certifications and Workshops

- Text To Intelligence: LLM Workshop, Vidyut 2025
- Improving Deep Neural Networks: Hyperparameter Tuning, Regularization and Optimization (Aug, 2025)
- Neural Networks and Deep Learning (Apr, 2025)
- Using Python to Interact with the Operating System by Google (Mar, 2025)
- Generative AI with Large Language Models by DeepLearning.AI (Dec, 2024)
- HTML, CSS, and Javascript for Web Developers by John Hopkins University (Aug, 2024)
- Microsoft Certified: Azure AI Fundamentals by Coincent (2023–2024)
- Python for Data Science, AI & Development by IBM (July, 2024)

Activities

- Volunteered at ICSRF (International Conference on Sustainable & Resilient Futures)
- Volunteered at ICTR-3 (International Conference on Tsunami Risk Reduction & Resilience) Seminar
- Volunteered at AmritaVarsham 70 & 71

Skills

Programming: Python, Java, C

Core CS: Data Structures and Algorithms, OOPS, DBMS

AI: Fundamentals of AI, Machine Learning, Deep Learning

Web Development: Flask, HTML, CSS , Fastapi

Database: MySQL, Postgres

Relevant CourseWork: OS, CN, IOT, Robotics, Bio Informatics

Tools: Jupyter Notebook, Matlab, Git

Soft skills: Problem-Solving, Team Collaboration, Communication, Critical Thinking, Flexible, Adaptive Learning